Foundation A

Marking guide • owner review

40 possible marks: 11 selected-response marks suitable for automatic checking; 29 written/working marks require structured self-marking or teacher review. These are original proposed schemes, not official AQA mark schemes.

Never award written scientific marks by keyword count. Accept equivalent valid science. Original submitted answers and review history must remain intact. A self-assessed total stays labelled self-assessed; a locally declared teacher mark is not authenticated external verification. No official GCSE grade follows from this topic paper.

1a · 1 mark · Selected response

Correct option: A. Cytoplasm

Cytoplasm.

Specification: 4.1.1.2 | Revision: O2

1b · 2 marks · Human/self-review

Nucleus (1); controls cell activities / contains genetic material or DNA (1). Accept either clear function as the second mark, not simply “important”.

Specification: 4.1.1.2 | Revision: O2

1c · 1 mark · Selected response

Correct option: B. A small ring of DNA

A small ring of DNA.

Specification: 4.1.1.1-4.1.1.2 | Revision: O2

1d · 1 mark · Selected response

Correct option: A. Plant cell

A plant cell; not every plant cell contains chloroplasts.

Specification: 4.1.1.1-4.1.1.2 | Revision: O2

2a · 1 mark · Selected response

Correct option: C. ×4

Start with ×4, the lowest-power objective.

Specification: 4.1.1.2 RP01 | Revision: O4/O5

2b · 1 mark · Human/self-review

Resolution / resolving power (1).

Specification: 4.1.1.5 | Revision: O4

2c · 3 marks · Human/self-review

24 ÷ 600 = 0.04 mm, or a correct equivalent rearrangement (1); valid mm-to-micrometre conversion, ×1000 (1); 40 µm (1). Fully correct answer with unit can earn 3. A correct calculation method with arithmetic error can retain method/conversion credit; do not give an answer mark for dimensional confusion.

Specification: 4.1.1.5; MS 1b/3b | Revision: O4/O5

2d · 1 mark · Human/self-review

The drawings have no calibrated scale/magnification, so displayed lengths cannot establish real lengths (1). Accept that the drawings may have different unknown magnifications. Do not credit simply saying that they look different.

Specification: 4.1.1.2 RP01; WS 3.7 | Revision: O5

3a · 1 mark · Selected response

Correct option: B. A chromosome is made of DNA and carries many genes

A chromosome is made of DNA and carries many genes.

Specification: 4.1.2.1 | Revision: O6

3b · 2 marks · Human/self-review

DNA replicates to make two copies of each chromosome (1); sets of chromosomes separate to opposite ends and nuclei divide (1). No named mitotic phases required.

Specification: 4.1.2.2 | Revision: O6

3c · 1 mark · Selected response

Correct option: C. Many plant cells can differentiate throughout life

Many plant cells can differentiate throughout life.

Specification: 4.1.1.4 | Revision: O3/O6

3d · 2 marks · Human/self-review

Mitochondria carry out aerobic respiration (1); this releases energy needed for the beating/movement of the projections (1). Accept respiration releases the energy for movement without the word aerobic. Do not award a mark just for repeating that there are many mitochondria. Energy is released, not created.

Specification: 4.1.1.3-4.1.1.4 | Revision: O3

4a · 2 marks · Human/self-review

A blood-cell type / blood cells (1); embryonic stem cells can form most types of human cell / a wider range than adult marrow (1).

Specification: 4.1.2.3 | Revision: O6

4b · 1 mark · Selected response

Correct option: C. An embryo with the same genes as the patient is produced

An embryo with the same genes as the patient is produced.

Specification: 4.1.2.3 | Revision: O6

4c · 2 marks · Human/self-review

Replacing damaged nerve cells could help restore lost function, supporting further research (1); viral transfer could harm a patient, or embryo use could make treatment unacceptable to a patient with a justified ethical/religious objection (1). Link the table evidence to a consequence or suitability judgement; simply copying two entries without a reason is insufficient. Do not require proof of effectiveness or imply a guaranteed cure.

Specification: 4.1.2.3; WS 1.3 | Revision: O6

5a · 1 mark · Selected response

Correct option: A. Net particle movement from higher to lower concentration

Net particle movement from higher to lower concentration.

Specification: 4.1.3.1 | Revision: O7

5b · 2 marks · Human/self-review

Valid surface-area/volume method: 6×2² and 2³, yielding 24 mm² and 8 mm³ if arithmetic is correct (1); correct ratio 3:1 or an equivalent ratio such as 24:8 (1). The question does not require simplest form, so a correct equivalent final ratio can earn 2. If earlier arithmetic is wrong, award at most the method mark; do not award full marks merely for simplifying a wrong ratio consistently. Do not credit area/volume dimensional confusion.

Specification: 4.1.3.1; MS 1c/5c | Revision: O8

5c · 2 marks · Human/self-review

Any two distinct feature-effect links (1 each): folds increase area; thin surface shortens diffusion distance; blood flow maintains concentration gradient. Simply naming features gets no explanation marks.

Specification: 4.1.3.1 | Revision: O8

5d · 1 mark · Selected response

Correct option: A. Diffusion was faster at 20 °C under these conditions

Diffusion was faster at 20 °C under these conditions.

Specification: 4.1.3.1; WS 3.5 | Revision: O7

6a · 2 marks · Human/self-review

Active transport moves ions from lower to higher concentration / against their gradient (1); requires energy from respiration (1).

Specification: 4.1.3.3 | Revision: O11

6b · 1 mark · Selected response

Correct option: A. Net movement of water from a dilute to a concentrated solution through a partially permeable membrane

Net movement of water from a dilute to a concentrated solution through a partially permeable membrane. Water, net direction and the partially permeable membrane are all part of the correct description.

Specification: 4.1.3.2 | Revision: O9

6c · 1 mark · Selected response

Correct option: A. A larger concentration difference

A larger concentration difference.

Specification: 4.1.3.1 | Revision: O7

7a · 2 marks · Human/self-review

(2.64 - 2.40) ÷ 2.40 × 100 (1); +10% (1). A fully correct signed final answer can earn 2 without written working. Award the method mark for a valid percentage-change method even if arithmetic goes wrong. A gain stated in words with 10% is acceptable; 0.24 g alone is not a percentage.

Specification: 4.1.3.2 RP02; MS 1c | Revision: O9/O10

7b · 6 marks · Human/self-review

Use the original level-of-response rubric in the blueprint: 0-6, human review. Indicative content: zero net mass change at about 0.40 for this tissue/conditions; positive/negative mass changes from net water movement across partially permeable membranes; single cylinders, surface liquid, different durations undermine comparison; equal dimensions/same tissue, equal time/temperature/solution volume, consistent gentle blotting, replicates and means, more concentrations near zero; do not generalise to all plant tissue or claim water stops moving. Explicitly credit comparing several plant tissue types across the concentration range under matched conditions (dimensions, time, temperature, solution volume and blotting), with repeats within each type and comparison of their estimated zero-change concentrations. Repeats of one tissue assess reliability but cannot establish the same zero-change concentration for every plant tissue; even agreement among several types cannot prove a universal claim.

Specification: 4.1.3.2 RP02; WS 2/3; MS 4a/4d | Revision: O9/O10

Six-mark best-fit rubric

0 marks: No relevant creditworthy content.

1-2 marks: A few relevant observations or improvements with little explanation; the judgement is weakly supported.

3-4 marks: Some linked use of the graph/osmosis and useful improvements, but the interpretation, justification or evaluation is incomplete.

5-6 marks: A coherent, evidence-based judgement links the zero-change estimate and net water movement to dataset limitations and a workable, controlled, repeated investigation that compares several tissue types under matched conditions and compares their zero-change estimates. Improvements are justified, and the limits of generalising to every plant tissue are explained.

Judge the whole response, select the best-fit level, then choose the lower/upper mark according to completeness and coherence. Do not count six keywords or require every possible control. For 5-6, both the scientific interpretation and justified methodological evaluation need to be developed. A strong answer can use other valid controls. For the top level, the response must develop how tissue-type comparisons under matched conditions address the universal claim; repeats of one tissue alone do not do so. Accept equivalent scientifically valid approaches and wording, including a justified counterexample design using another tissue type. Do not require a set phrase, named tissues, every listed control, or a checklist of points; apply the levels holistically. Merely saying “repeat for accuracy” is less developed than explaining that repeats expose variation and support a mean.

Illustrative model, not required wording

For this celeriac tissue under these conditions, the graph suggests no net mass change at about 0.40 mol/dm³. Below this concentration the tissue gains mass as water enters overall by osmosis through partially permeable cell membranes; above it the tissue loses water overall. This does not establish the same value for every plant tissue. Surface liquid would affect the measured mass, and different soaking times make comparisons unfair. I would use equal-sized cylinders from comparable tissue, equal solution volumes, time and temperature, and consistent gentle blotting before weighing. Repeats and means would improve the evidence, with extra concentrations near 0.40 to refine the estimate. I would also test several different plant tissue types using the same concentration range and matched conditions, repeating each type and comparing their estimated zero-change concentrations. Different estimates would refute the claim; similar estimates for the tested types would still not prove it for every plant tissue. No net change does not mean water molecules stop moving.

Foundation graph estimate: approximately 0.40 mol/dm³. Higher: approximately 0.32 mol/dm³; allow a justified graphical estimate from 0.31 to 0.33 on the supplied graph. These are estimates, not required spurious exactness. Percent changes in 7a are from separate pilot batches and are not answered by the 7b graph.